Diffusion Models: Methodology and Applications

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Statement of Contribution

This section outlines the specific contributions of each group member to this report:

- **Chris:** Background, Experiments, Discussion and Conclusion, and proofreading.
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1 Introduction

Generative modelling refers to using machine learning techniques to learn a data distribution so that new data points can be sampled from that distribution. The main applications of generative modelling include image and video generation, music composition, and text generation. This report will focus on learning an image distribution with the focus on sampling novel images from the distribution.

1.1 Generative Models

The variational auto-encoder (VAE) was proposed in 2013 (Kingma and Welling, 2013), and improved how efficiently images could be sampled. The VAE's goal is to use an encoder to represent a complex data distribution as a lower-dimensional manifold, and then use a decoder to recreate the initial data distribution. The VAE's loss objective and encoder force the manifold space to take the shape of a normal multivariate distribution. While prior sampling methods required iteratively searching a sparse manifold to find the correct latent to input into the decoder so it can create the images, the VAE method (Kingma and Welling, 2013) only needs to use the encoder once to compute the mean of the manifold; then, by slightly adjusting the mean, the decoder can sample novel generations from the data distribution.

In 2014, the generative adversarial network (GAN) was proposed as a new framework for generating images (Goodfellow et al., 2014), and showed the potential for GANs, especially in generating sharp images compared to prior approaches. The GAN model uses two components: the discriminator and the generator. The generator is trained to create the images, whilst the discriminator is trained to distinguish between the real and generated images. The generator uses the loss function defined by the discriminator's predictions to produce images of higher quality. Since 2014, GANs have continued to be the main focus of generative modelling; state-of-the-art networks can make very high-quality images (Bermano et al., 2022). Generative adversarial networks can suffer from mode collapse (Goodfellow et al., 2014), where the generator finds a single high-quality output that can trick the discriminator, leading to the generator not exploring other areas of the data distribution.

Diffusion models are a category of generative models that involve adding noise to an image and learning to reverse that process. The denoising diffusion probabilistic model (DDPM) is a type of diffusion model that was shown to be a viable method for generative modelling (Ho et al., 2020). The DDPM is trained by noising an image from the data distribution by random amounts and training a neural network to predict the amount of noise added to the clean image. The DDPM then uses this trained neural network to iteratively predict the noise and transform Gaussian noise into a high-quality image. Random noise is also added at each step so the DDPM can sample diverse images.

The **flow matching model** is a type of generative model that constructs a probability path that transforms a base distribution into the data distribution (Lipman et al., 2022). Instead of directly learning the probability path, a time-dependent velocity field is used. An ODE solver is then used to iteratively integrate over the velocity field and sample from the data distribution. The velocity field is not known, so it must be approximated by a neural network. The flow matching loss objective uses the mean square to regress the real and predicted velocity fields. The simplest velocity field is called the optimal transport (OT), which is defined by a straight-line trajectory between a Gaussian noise distribution and the data distribution. The OT is helpful due to the simplicity of the true velocity field.

1.2 Report Motivation

The DDPM was shown to be a promising model for image generation, as evidenced by it outperforming the 2021 state-of-the-art models like GANs on unconditional image generation after a series of ablations (Dhariwal and Nichol, 2021). Close variants to the DDPM, like the denoising diffusion implicit models (DDIM), can close the gap of sampling efficiency by producing high-quality samples 10 to 50 times faster (Song et al., 2021). The DDPM also addresses many issues that other models have, like the lack of diversity in GANs due to their adversarial nature of training. Overall, the primary motivation for this report is to provide the reader with the necessary fundamental knowledge of DDPMs to be built upon in further reports that talk about a more domain-specific area of diffusion models.

1.3 Report Structure

In this report, we first introduce the theoretical foundations of the DDPM, and then the implementation of the models and the evaluation of the performance. The background (Section 2) will mathematically explore the main features of the DDPM, and give insight into four other diffusion model variants. The methodology (Section 3) will explain the design choices made during the implementation of the diffusion model and the steps made to ensure the statistical reliability of the results presented. The experiments (Section 4) will provide quantitative metrics and qualitative images of two different levels of dataset difficulty to give multiple different angles of the effectiveness of the DDPM. The discussion and conclusion (Section 5) will reflect on the DDPM and discuss how other diffusion model variants fix various problems the standard DDPM has.

2 Background

This section establishes the mathematical and architectural foundations required to rigorously analyse and evaluate Denoising Diffusion Probabilistic Models (DDPMs).

Problem Formulation. The fundamental objective of generative modelling is to approximate an unknown, complex ground-truth data distribution $q_0(x_0)$ using a tractable, parameterised model distribution $p_\theta(x_0)$. In the context of diffusion models, this problem is formulated as a two-stage stochastic process: a forward corruption process that systematically maps $q_0(x_0)$ to a predefined prior distribution (typically a standard Gaussian), and a reverse denoising process parameterised by θ that learns to invert this mapping. To systematically verify and extend the generative capabilities of DDPMs across diverse data manifolds, it is imperative to first establish the underlying probabilistic bounds and the corresponding parameterised transition dynamics.

Structure. The remainder of this background is organised as follows to provide a comprehensive theoretical framework: First, Section 2.1 introduces the foundational DDPM framework, detailing the continuous state-space Markovian transitions and the variational training objective. Building upon this baseline, Section 2.2 reviews key theoretical extensions (Variants), encompassing noise schedule optimisation, non-Markovian accelerated sampling, discrete categorical state spaces, and conditional generative guidance. Subsequently, Section 2.3 defines the mathematical topologies of the neural network architectures (MLP and U-Net) utilised to parameterise the noise predictors. Finally, Section 2.4 establishes the empirical evaluation metrics, constructing a symmetric diagnostic framework to quantify distribution alignment and mode coverage across both low-dimensional geometric manifolds and high-dimensional semantic spaces.

Notations. Throughout this report, standard lowercase letters (e.g., t, x, y) denote both scalars and state-space vectors, while boldface characters (e.g., the zero vector $\mathbf{0}$ and identity matrix $\mathbf{I}$) represent constant vectors and matrices. Specifically, $q_0(x_0)$ denotes the **original real data distribution**. The terms $q_t(x_t)$ and $p_t(x_t)$ represent the marginal densities of the **true forward process** and the **generative reverse process** at time $t \in \{0, \dots, T\}$, respectively. Additionally, $p_\theta(\cdot)$ denotes probability distributions parameterised by a neural network with learnable weights $\theta \in \Theta$ (e.g., the reverse transition probability $p_\theta(x_{t-1} \mid x_t)$). We use $\nabla_{x_t} f(x_t)$ to signify the gradient of a function $f(x_t)$ with respect to the state variable x_t. Furthermore, $\mathbb{E}_{x\sim p}[\cdot]$ denotes the mathematical expectation under distribution p, $\mathcal{N}(\mu, \Sigma)$ represents a Gaussian distribution with mean μ and covariance Σ, and $\mathcal{U}(S)$ designates a uniform distribution over a given set or interval S. Symbols $\|\cdot\|$ and $|\cdot|$ denote the L_2 norm and absolute value, $\sim$ indicates "distributed as", and $\triangleq$ or $:=$ signify "defined as".

2.1 Diffusion Model Foundations

2.1.1 Generation Theory

The main idea behind the standard diffusion model DDPM is to progressively corrupt the data $x_0 \sim q_0(x_0)$ and subsequently learn to reconstruct it, thereby accomplishing the generative task (Sohl-Dickstein et al., 2015). Specifically, the diffusion model comprises two distinct processes: first, the **forward noising process** q, which gradually injects Gaussian noise into clean image data x_0 over T steps until it loses all distinguishable features, reducing the data to a pure random noise state x_T; and second, the **reverse denoising process** p_θ, where a parameterised neural network is trained to predict and iteratively remove the added noise, ultimately recovering a clear image x_0 from the pure noise prior $p_T(x_T)$ (Ho et al., 2020). To formulate the generative training objective for this noise prediction, this section first establishes the mathematical definition of the forward noising process.

Definition 1 (Forward Process of DDPM Ho et al. (2020))**.** Given an initial data state $x_0 \in \mathbb{R}^d$ following the real data distribution $x_0 \sim q_0(x_0)$, the forward process of the DDPM is defined as a discrete-time Markov chain. For discrete time steps $t \in \{1, 2, \ldots, T\}$, given a monotonically increasing variance schedule $\{\beta_t\}_{t=1}^T$ with $\beta_t \in (0, 1)$, and defining the state sequence $x_{1:T} \triangleq \{x_1, \ldots, x_T\}$, the joint probability density of the forward process is given by:

$$q_{1:T}(x_{1:T} \mid x_0) = \prod_{t=1}^{T} q_t(x_t \mid x_{t-1})$$

where the single-step transition probability density from data state x_{t-1} to x_t is defined as a Gaussian distribution:

$$q_t(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t \mathbf{I})$$

Its equivalent recursive form in terms of random variables is:

$$x_t = \sqrt{1-\beta_t}x_{t-1} + \sqrt{\beta_t}\epsilon_t$$

where $\epsilon_t \in \mathbb{R}^d$ is standard Gaussian noise independently sampled at each time step, satisfying $\epsilon_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and $\mathbf{I} \in \mathbb{R}^{d\times d}$ denotes the identity matrix.

Based on the single-step forward transition $q_t(x_t \mid x_{t-1})$ defined above, obtaining the noisy image x_t at any timestep t necessitates sequential iterations starting from the initial data x_0. In practical applications, since the total number of timesteps T in standard diffusion models is typically substantial (e.g., $T = 500$), this iterative procedure renders training inefficient. To address this computational bottleneck, we employ the reparameterisation trick, which enables the derivation of a closed-form marginal distribution $q_t(x_t \mid x_0)$ for the multi-step forward process directly dependent on the initial state x_0. This allows for the direct sampling of noisy images at any arbitrary timestep t, thereby significantly improving training efficiency.

Proposition 1 (Gaussian Marginal Forward Distribution)**.** Based on the Markov chain described in Definition 1, let the single-step mean scaling factor be $\alpha_t := 1 - \beta_t$ and the cumulative scaling factor be $\bar{\alpha}_t := \prod_{i=1}^{t} \alpha_i$. Then, for any time step t, given the initial state x_0, the marginal forward distribution (transition distribution) can be strictly derived as:

$$q_t(x_t \mid x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)\mathbf{I})$$

Its equivalent representation in terms of random variables is:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$$

where $\epsilon \in \mathbb{R}^d$ is the equivalent standard Gaussian noise aggregating all single-step noises from step 1 to t, satisfying $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

Proof. The detailed mathematical proof is deferred to Appendix A.1. □

While the aforementioned formulations enable data corruption, the converse objective of diffusion models is to iteratively remove noise starting from a pure noise state $x_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, ultimately recovering a clean image x_0 that follows the true data distribution $q_0(x_0)$. Theoretically, this necessitates calculating the true reverse transition probability $q_t(x_{t-1} \mid x_t)$. However, directly estimating this distribution requires marginalisation over the entire data distribution space, which is mathematically intractable (Sohl-Dickstein et al., 2015). Nevertheless, by conditioning on the initial image x_0, the posterior distribution $q_t(x_{t-1} \mid x_t, x_0)$ can be expressed in Gaussian form, signifying that the true reverse trajectory possesses an exact analytical expression.

Proposition 2 (Conditional Posterior of the Forward Process)**.** Consistent with the Markov chain transitions in Definition 1 and the forward marginal distribution in Proposition 1, given the initial data state x_0, the conditional posterior distribution $q_t(x_{t-1} \mid x_t, x_0)$ of the system state x_{t-1} conditioned on x_t is Gaussian, and its analytical form strictly satisfies:

$$q_t(x_{t-1} \mid x_t, x_0) = \mathcal{N}(x_{t-1}; \tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t \mathbf{I})$$

where the posterior mean vector $\tilde{\mu}_t(x_t, x_0) \in \mathbb{R}^d$ and the posterior variance scalar $\tilde{\beta}_t \in (0, 1)$ are defined as:

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} x_0 + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t$$

$$\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}\beta_t$$

Proof. The detailed mathematical proof is deferred to Appendix A.2. □

The aforementioned closed-form expressions indicate that the variance of the true posterior distribution, $\tilde{\beta}_t$, is a constant depending solely on pre-defined noise scheduling hyperparameters β_t and $\bar{\alpha}_t$ (i.e., $\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$). This implies that when utilising a parameterised neural network $p_\theta(x_{t-1} \mid x_t)$ to approximate this true posterior $q_t(x_{t-1} \mid x_t, x_0)$, the network is not required to predict the variance σ_t^2; its primary objective is to accurately predict the distribution mean $\tilde{\mu}_t(x_t, x_0)$.

Definition 2 (Reverse Process of DDPM Ho et al. (2020))**.** Consistent with the forward process established in Definition 1, given the final discrete timestep T, the reverse process of DDPM is defined as a Markov chain parameterised by learnable parameters $\theta \in \Theta$. This process originates from a prior distribution, strictly defined as a standard Gaussian distribution:

$$p_T(x_T) = \mathcal{N}(x_T; \mathbf{0}, \mathbf{I})$$

For discrete timesteps $t \in \{T, T-1, \ldots, 1\}$, the joint probability density distribution $p_\theta(x_{0:T})$ of the reverse process factorises as:

$$p_\theta(x_{0:T}) = p_T(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t)$$

Specifically, the single-step reverse transition probability $p_\theta(x_{t-1} \mid x_t)$ from state x_t to x_{t-1} is parameterised as a Gaussian distribution:

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_t^2 \mathbf{I})$$

where $\mu_\theta(x_t, t) \in \mathbb{R}^d$ denotes the mean vector of the reverse transition predicted by a neural network, and $\sigma_t \in \mathbb{R}^+$ represents the standard deviation parameter corresponding to this discrete reverse transition.

2.1.2 Training Objective

In probabilistic generative modelling, a fundamental objective is to train the model such that it assigns higher probabilities to samples drawn from the true data distribution $q_0(x_0)$. Mathematically, this corresponds to maximising the expected log-likelihood: $\max \mathbb{E}_{x_0 \sim q_0(x_0)}[\log p_\theta(x_0)]$. However, reconstructing a clean image x_0 from pure noise x_T involves a sequence of latent variables across multiple diffusion steps. Calculating the marginal likelihood $p_\theta(x_0)$ necessitates marginalisation over all possible intermediate states $x_{1:T}$, which is computationally intractable in high-dimensional spaces (Sohl-Dickstein et al., 2015).

To circumvent this computational bottleneck, diffusion models employ a variational inference framework (Sohl-Dickstein et al., 2015). Rather than directly maximising the intractable log-likelihood $\log p_\theta(x_0)$, the objective is reformulated to maximise a tractable surrogate known as the Evidence Lower Bound

(ELBO), denoted as L_{VLB} . Because the ELBO constitutes a strict lower bound, maximising L_{VLB} indirectly optimises the true log-likelihood of generating authentic data x_0 (Ho et al., 2020).

Theorem 1 (Variational Lower Bound of DDPM Ho et al. (2020))**.** Consistent with the forward process defined in Definition 1 and the reverse process in Definition 2, given the initial real data state $x_0 \sim q_0(x_0)$, the optimisation objective for model parameters $\theta \in \Theta$ is to minimise the negative log-likelihood $-\log p_\theta(x_0)$. This objective is strictly bounded above by the negative Evidence Lower Bound (ELBO), denoted as L_{VLB}, such that $-\log p_\theta(x_0) \leq L_{VLB}$. This bound can be rigorously decomposed into three independent components:

$$L_{VLB} = \mathbb{E}_{q_0(x_0)} \left[L_T + \sum_{t=2}^{T} L_{t-1} + L_0 \right]$$

where each constituent component is formally defined as:

$$L_T = D_{KL}(q_T(x_T \mid x_0) \parallel p_T(x_T))$$

$$L_{t-1} = \mathbb{E}_{q_t(x_t|x_0)} \left[D_{KL}(q_t(x_{t-1} \mid x_t, x_0) \parallel p_\theta(x_{t-1} \mid x_t)) \right]$$

$$L_0 = -\mathbb{E}_{q_1(x_1|x_0)} [\log p_\theta(x_0 \mid x_1)]$$

Specifically, the intermediate transition loss L_{t-1} is analytically equivalent to:

$$L_{t-1} = \mathbb{E}_{q_t} \left[\frac{1}{2\sigma_t^2} \|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\|^2 \right] + C$$

where C is a constant independent of the network parameters θ, and $D_{KL}(P \parallel Q)$ denotes the Kullback-Leibler divergence between probability distributions P and Q.

Through algebraic expansion, the objective of maximising the ELBO reduces to minimising a series of Kullback-Leibler (KL) divergence terms L_{t-1}. In this context, the KL divergence quantifies the discrepancy between the learned reverse transition $p_\theta(x_{t-1} \mid x_t)$ and the true forward posterior $q_t(x_{t-1} \mid x_t, x_0)$. Consequently, minimising these divergences drives the learned reverse process p_θ to approximate the true reverse diffusion dynamics q_t. Within the DDPM formulation, both distributions are modelled as Gaussians with fixed variances (σ_t^2 and $\tilde{\beta}_t$). Thus, aligning these two distributions reduces primarily to matching their respective means, $\mu_\theta(x_t, t)$ and $\tilde{\mu}_t(x_t, x_0)$ (Ho et al., 2020). To render this mean-matching objective computationally tractable for neural network training, we demonstrate that by reparameterising the true posterior mean $\tilde{\mu}_t$, the network's objective fundamentally simplifies to estimating the injected noise ϵ.

Proposition 3 (Reparameterisation of the Posterior Mean Ho et al. (2020))**.** Building upon the conditional posterior mean $\tilde{\mu}_t(x_t, x_0)$ derived in Proposition 2, the true conditional posterior mean can be rigorously reformulated as a

function of the current state x_t and the standard Gaussian noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$:

$$\tilde{\mu}_t(x_t, x_0) = \tilde{\mu}_t(x_t, \epsilon) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon \right)$$

Leveraging this structural equivalence, we introduce a parameterised mapping function $\epsilon_\theta : \mathbb{R}^d \times \{1, \ldots, T\} \to \mathbb{R}^d$ to define the reverse transition mean $\mu_\theta(x_t, t)$ from Definition 2 as:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right)$$

Proof. The detailed mathematical proof for this reparameterisation is deferred to Appendix A.3. □

This transformation reveals a crucial engineering insight (Ho et al., 2020): since the state x_t acts as an accessible input to the network and $\alpha_t, \bar{\alpha}_t$ are predefined constants, predicting the mean of the reverse distribution $\mu_\theta(x_t, t)$ is mathematically equivalent to predicting the exact noise ϵ injected at timestep t . Consequently, instead of training the network to directly output the high-dimensional image mean, we train a noise prediction network $\epsilon_\theta(x_t, t)$. Furthermore, by substituting this noise parameterisation into the variational objective L_{t-1}, the optimisation function of the ELBO reduces to a straightforward regression problem.

Proposition 4 (Equivalence of Stepwise ELBO)**.** Following the stepwise loss term L_{t-1} established in Theorem 1 and setting the reverse transition variance parameter to $\sigma_t^2 = \tilde{\beta}_t$, substituting the analytically derived $\tilde{\mu}_t(x_t, \epsilon)$ and $\mu_\theta(x_t, t)$ from Proposition 3 into L_{t-1} demonstrates that this loss term is strictly equivalent to a weighted quadratic form with respect to the noise residual:

$$L_{t-1} = \mathbb{E}_{x_0, \epsilon} \left[\frac{\beta_t^2}{2\sigma_t^2 \alpha_t (1-\bar{\alpha}_t)} \left\| \epsilon - \epsilon_\theta(x_t, t) \right\|^2 \right] + C$$

where the constant C is independent of the network parameters θ.

Proof. The detailed mathematical proof is deferred to Appendix A.4. □

Following these analytical transformations, we can now give the formal definition of the simplified objective function $L_{simple}(\theta)$ for DDPM :

Definition 3 (Simplified Training Objective of DDPM Ho et al. (2020))**.** Based on the equivalent stepwise variational lower bound derived in Proposition 4, the final parameterised training objective for the DDPM is formally defined as an unweighted functional $L_{simple}(\theta)$. For a given set of learnable neural network parameters $\theta \in \Theta$, this objective functional is defined as:

$$L_{simple}(\theta) = \mathbb{E}_{t, x_0, \epsilon} \left[\left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t} \epsilon, t) \right\|^2 \right]$$

which can be equivalently denoted as:

$$L_{simple}(\theta) \equiv \mathbb{E}_{t,x_t,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right]$$

where the intermediate state follows the transition $x_t = \sqrt{1-\beta_t}x_{t-1} + \sqrt{\beta_t}\epsilon_t$, the initial data distribution is $x_0 \sim q_0(x_0)$, the discrete timestep is uniformly distributed such that $t \sim \mathcal{U}(\{1, \ldots, T\})$, and the injected noise is standard Gaussian $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

Algorithm 1 DDPM Training Procedure Ho et al. (2020)

1: **repeat**
2: $x_0 \sim q_0(x_0)$
3: $t \sim \mathcal{U}(\{1, \ldots, T\})$
4: $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on

$$\nabla_\theta \left\|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t)\right\|^2$$

6: **until** converged

2.1.3 Sampling Rule

Once trained, the parameterised neural network $\epsilon_\theta(x_t, t)$ is capable of predicting the noise component inherent within a noisy state x_t. Leveraging this capability, diffusion models generate novel samples by simulating the learned reverse diffusion process p_θ, progressively transforming a pure random noise state x_T into structured data x_0 that aligns with the original real data distribution $q_0(x_0)$. Within the DDPM framework, this generative phase is conventionally termed sampling or reconstruction (Ho et al., 2020). The specific sampling rule is defined as follows:

Definition 4 (DDPM Sampling Rule)**.** Consistent with the forward process established in Definition 1, for reverse iterative timesteps $t \in \{T, T-1, \ldots, 1\}$, given a trained noise predictor $\epsilon_\theta(x_t, t) \in \mathbb{R}^d$, the single-step reverse sampling process of the unconditional DDPM is defined as:

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z_t$$

where the parameterised mean vector $\mu_\theta(x_t, t) \in \mathbb{R}^d$ is formulated as:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right)$$

Here, the scalar $\sigma_t > 0$ denotes the standard deviation parameter corresponding to the discrete reverse transition distribution. Furthermore, $z_t \in \mathbb{R}^d$ represents

standard Gaussian noise sampled independently, satisfying $z_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ for $t > 1$, and $z_1 = \mathbf{0}$. The schedule parameters α_t and $\bar{\alpha}_t$ are strictly inherited from Proposition 1.

To execute this generative phase, the sampling procedure operates as a reverse Markov chain trajectory:

$$x_T \to x_{T-1} \to \cdots \to x_0$$

Practically, the model initiates the process by drawing a prior state x_T from a standard isotropic Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. At each subsequent discrete timestep t, the network $\epsilon_\theta(x_t, t)$ estimates the inherent noise to compute the specific mean parameter $\mu_\theta(x_t, t)$ of the reverse transition distribution $p_\theta(x_{t-1} \mid x_t)$, from which the subsequent less-noisy state x_{t-1} is drawn. This progressive denoising algorithm, which ultimately yields the generated sample x_0, is summarised in Algorithm 2 .

Algorithm 2 DDPM Sampling Procedure Ho et al. (2020)

1: $x_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, T-1, \ldots, 1$ **do**
3: $\quad z_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $z_1 = \mathbf{0}$
4: $\quad x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z_t$
5: **end for**
6: **return** x_0

2.2 Theoretical Extensions and Variants

2.2.1 Extension to Learnable Variances

In the foundational DDPM framework, the forward process typically employs a linear variance schedule, and the reverse process variance is restricted to a non-learnable constant (Ho et al., 2020). However, this configuration presents two notable limitations: a linear schedule destroys macroscopic structural information too rapidly at early timesteps, and a fixed variance restricts the model's capacity to fit the true generation distribution when the number of sampling steps is compressed. To overcome these constraints, Nichol and Dhariwal (2021) proposed the **Improved DDPM framework**, which parameterises the model with a distinct set of learnable weights $\phi \in \Phi$ to differentiate it from the baseline DDPM. This framework introduces a cosine noise schedule, governed by a new set of schedule parameters $\bar{\alpha}'_t$ and β'_t, alongside an adaptive, parameterised variance $\Sigma_\phi(x_t, t)$.

Definition 5 (Cosine Noise Schedule Nichol and Dhariwal (2021))**.** Given the total discrete timesteps T and a small constant offset $s > 0$ (typically $s = 0.008$), the cumulative scaling factor $\bar{\alpha}'_t$ for any discrete timestep $t \in \{0, 1, \ldots, T\}$ under the cosine schedule is analytically defined as:

$$\bar{\alpha}'_t = \frac{f(t)}{f(0)}, \quad \text{where} \quad f(t) = \cos^2\left(\frac{t/T + s}{1+s} \cdot \frac{\pi}{2}\right)$$

The corresponding single-step variance β'_t is derived as $\beta'_t = 1 - \frac{\bar{\alpha}'_t}{\bar{\alpha}'_{t-1}}$, which is clipped at $\beta'_t = \min(\beta'_t, 0.999)$ to preserve numerical stability.

To prevent premature information loss during the forward process, this cosine formulation ensures a smoother change rate compared to the linear schedule, allowing data features to be corrupted uniformly across the entire T-step sequence. Furthermore, to address the posterior distribution widening caused by accelerated sampling, Improved DDPM upgrades the fixed variance to an adaptive parameter $\Sigma_\phi(x_t, t)$ driven by the network weights ϕ.

Definition 6 (Adaptive Variance of Improved DDPM Nichol and Dhariwal (2021))**.** Relaxing the constraint of a constant variance in the unconditional DDPM, the reverse transition distribution $p_\phi(x_{t-1} \mid x_t)$ is parameterised by the weights $\phi \in \Phi$. A non-linear mapping function $v_\phi : \mathbb{R}^d \times \{1, \ldots, T\} \to [0, 1]^d$ is introduced. The reverse covariance matrix $\Sigma_\phi(x_t, t)$ is formulated as an element-wise convex interpolation in the logarithmic space between the untrained constants β'_t and $\tilde{\beta}'_t$:

$$\Sigma_\phi(x_t, t) = \text{diag}\left(\exp\left(v_\phi(x_t, t) \log \beta'_t + (\mathbf{1} - v_\phi(x_t, t)) \log \tilde{\beta}'_t\right)\right)$$

where the constant $\tilde{\beta}'_t = \frac{1-\bar{\alpha}'_{t-1}}{1-\bar{\alpha}'_t}\beta'_t$, and $\mathbf{1} \in \mathbb{R}^d$ represents a vector of ones. Both the exponential and logarithmic functions operate element-wise.

Learning the interpolation vector v_ϕ allows the generative model p_ϕ to dynamically adjust the posterior distribution width. In summary, by integrating the smooth cosine schedule $\bar{\alpha}'_t$ and the adaptive variance Σ_ϕ, Improved DDPM achieves a substantial acceleration in generation efficiency. It reduces the required reverse sampling steps by an order of magnitude (e.g., from 1000 to 50 steps) while maintaining or exceeding the high-fidelity sample quality of the standard DDPM (Nichol and Dhariwal, 2021).

2.2.2 Extension to Non-Markovian Dynamics

In the standard DDPM framework, the generative process is formulated as a strict reverse Markov chain (Ho et al., 2020). While this mathematical design ensures training stability, it introduces a substantial computational bottleneck

during inference: generating a single sample requires sequential iteration over all T timesteps. Because each step explicitly depends on the immediate preceding state, skipping steps is mathematically prohibited under the Markov assumption, rendering the sampling process excessively slow for real-time applications.

To overcome this limitation, Song et al. (2021) proposed **Denoising Diffusion Implicit Models (DDIM)**. The core innovation of DDIM lies in the realisation that the simplified training objective $L_{simple}(\theta)$ established in Definition 3 depends solely on the marginal distribution $q_t(x_t \mid x_0)$ in Proposition 1, rather than the specific joint distribution of the entire Markov chain. Leveraging this insight, DDIM constructs a family of non-Markovian forward processes that yield marginal distributions identical to those of DDPM but offer substantial flexibility in designing reverse sampling trajectories $p_\sigma(x_{t-1} \mid x_t, x_0)$.

Definition 7 (DDIM Sampling Rule Song et al. (2021))**.** Consider a reverse transition distribution $p_\sigma(x_{t-1} \mid x_t, x_0)$ for the discrete timesteps $t \in \{T, T-1, \ldots, 1\}$. Given a trained DDPM noise prediction network $\epsilon_\theta(x_t, t) \in \mathbb{R}^d$ parameterised by the baseline weights $\theta \in \Theta$, and a variance control parameter $\sigma_t \in \mathbb{R}_{\geq 0}$ satisfying $\sigma_t^2 \leq 1 - \bar{\alpha}_{t-1}$, the single-step sampling mapping for states $x_t, x_{t-1} \in \mathbb{R}^d$ is defined as:

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \left(\frac{x_t - \sqrt{1-\bar{\alpha}_t}\epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}} \right) + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}\epsilon_\theta(x_t, t) + \sigma_t z_t$$

where $z_t \in \mathbb{R}^d$ is a standard Gaussian random variable $z_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ sampled independently of x_t. For the final step $t = 1$, we define $z_1 = \mathbf{0}$, with the boundary condition $\bar{\alpha}_0 \triangleq 1$; The linear schedule parameters $\{\alpha_t, \beta_t, \bar{\alpha}_t\}_{t=1}^T$ directly from Definition 1 and Proposition 1..

In Definition 7, the first term in the summation represents the "predicted x_0" component, which provides the deterministic direction of the trajectory. This generalised formulation mathematically unifies stochastic and deterministic sampling via the adjustment of σ_t. By explicitly setting the variance parameter to zero ($\sigma_t = 0$), the random noise term $\sigma_t z_t$ is entirely eliminated, translating the reverse sampling into solving an empirical ordinary differential equation (ODE) (Song et al., 2021). Conversely, setting $\sigma_t = \sqrt{(1-\bar{\alpha}_{t-1})(1-\alpha_t)/(1-\bar{\alpha}_t)}$ strictly recovers the standard Markovian DDPM sampling defined in Definition 4.

Crucially, because this ODE-based numerical integration permits larger step sizes without violating the marginal probabilistic assumptions $q_t(x_t \mid x_0)$, the sampling trajectory is no longer restricted to continuous timesteps. Generating samples over a significantly shorter sub-sequence $\tau \subset \{1, \ldots, T\}$ accelerates the process by an order of magnitude. Ultimately, DDIM resolves the foundational speed bottleneck of diffusion models and can be directly applied to any pre-trained DDPM network without requiring further parameter updates (Song et al., 2021).

2.2.3 Extension to Discrete State Spaces

In the standard DDPM framework, the forward corruption process is strictly defined by injecting additive continuous Gaussian noise (Ho et al., 2020). This fundamental formulation restricts the model's applicability to continuous state spaces. However, many real-world datasets, such as natural language text or biological sequences, are inherently discrete. Applying continuous noise to categorical data is mathematically undefined. To address this theoretical limitation, Austin et al. (2021) introduced **Discrete Denoising Diffusion Probabilistic Models (D3PM)**, which reconstruct the forward process using discrete Markov transition matrices Q_t. To distinguish this categorical generative process from the continuous baseline, D3PM is parameterised by a distinct set of learnable weights $\psi \in \Psi$, and the discrete system state is denoted as x'_t.

Definition 8 (Discrete Forward Process of D3PM Austin et al. (2021))**.** Let the discrete system state x'_t be a random variable with K mutually exclusive categories, represented as a one-hot row vector $x'_t \in \{0,1\}^K$ such that $\mathbf{1}^T x'_t = 1$. Introducing a single-step transition matrix $Q_t \in [0,1]^{K\times K}$ that satisfies the row-stochastic property, the forward Markov chain of D3PM is defined via a Categorical distribution:

$$q(x'_t \mid x'_{t-1}) = \mathrm{Cat}(x'_t; x'_{t-1}Q_t)$$

Correspondingly, given the initial data x'_0, the forward marginal distribution at any discrete timestep t is strictly equivalent to:

$$q(x'_t \mid x'_0) = \mathrm{Cat}(x'_t; x'_0\bar{Q}_t)$$

where $\bar{Q}_t = Q_1 Q_2 \dots Q_t$ denotes the cumulative transition matrix.

Because the forward process is governed by explicitly defined transition matrices Q_t, the true discrete reverse posterior $q(x'_{t-1} \mid x'_t, x'_0)$ remains analytically tractable via Bayes' theorem (Austin et al., 2021).

Proposition 5 (Categorical Posterior of D3PM Austin et al. (2021))**.** Based on the discrete transition probabilities in Definition 8, given the initial data state x'_0, the true conditional posterior distribution $q(x'_{t-1} \mid x'_t, x'_0)$ also follows a Categorical distribution, with its analytical probability vector given by:

$$q(x'_{t-1} \mid x'_t, x'_0) = \mathrm{Cat}\left(x'_{t-1}; \frac{x'_t Q_t^T \odot x'_0 \bar{Q}_{t-1}}{x'_0 \bar{Q}_t {x'_t}^T}\right)$$

where $\odot$ denotes the element-wise Hadamard product, Q_t^T is the transpose of Q_t, and the denominator $x'_0 \bar{Q}_t {x'_t}^T$ serves as a normalising scalar.

In D3PM, rather than predicting a continuous noise vector governed by θ, the generative reverse process is parameterised by the distinct weights ψ. The neural network is trained to output categorical probabilities $p_\psi(\tilde{x}'_0 \mid x'_t)$ to predict

the uncorrupted token x'_0. By substituting this predicted initial state into the analytical posterior $q(x'_{t-1} \mid x'_t, x'_0)$, the model systematically constructs the reverse transition $p_\psi(x'_{t-1} \mid x'_t)$ (Austin et al., 2021). Ultimately, by substituting continuous Gaussian noise with discrete stochastic transition matrices Q_t, D3PM resolves the incompatibility between standard diffusion models and categorical data, bridging the gap between continuous and discrete generative modelling.

2.2.4 Extension to Conditional Guidance

In the foundational DDPM framework, the reverse process strictly approximates an unconditional data distribution $p_\theta(x_0)$ (Ho et al., 2020). This sampling process is fundamentally uncontrollable, posing a limitation in high-stakes domains like medical imaging, where practitioners require targeted synthesis or segmentation based on specific anatomical conditions. To overcome this, guided diffusion models introduce a condition variable $y \in \mathcal{Y}$ to shift the generative target to a conditional distribution $p_\omega(x_0 \mid y)$. To strictly distinguish this conditional network from the foundational unconditional model, it is parameterised by a distinct set of learnable weights $\omega \in \Omega$. A prominent application of this paradigm in the recent literature is **MedSegDiff** (Wu et al., 2022), which formulates medical image segmentation as a conditional diffusion process, guiding the denoising trajectory using the original medical scans as the condition y.

The theoretical cornerstone enabling this precise spatial control without requiring an independently trained external classifier is Classifier-Free Guidance (CFG) (Ho and Salimans, 2021). Rather than relying on auxiliary classification models, CFG jointly trains the parameterised noise prediction network $\epsilon_\omega(x_t, t, y)$ on both conditional and unconditional objectives by randomly dropping the condition y (replacing it with a predefined null token $\varnothing$). During inference, the generation is mathematically steered by linearly extrapolating between the conditional noise estimate $\epsilon_\omega(x_t, t, y)$ and the unconditional noise estimate $\epsilon_\omega(x_t, t, \varnothing)$. Specifically, the final adjusted noise estimate is computed as

$$\tilde{\epsilon}_\omega(x_t, t, y) = (1 + g)\epsilon_\omega(x_t, t, y) - g\epsilon_\omega(x_t, t, \varnothing)$$

where the scalar $g \geq 0$ designates the guidance scale.

This mathematical extrapolation explicitly pushes the probability mass away from the generic unconditional data manifold and concentrates it towards the specific region satisfying the condition y (Ho and Salimans, 2021). Ultimately, by reformulating the reverse trajectory into a controllable process via CFG, variants like MedSegDiff resolve the uncontrollability bottleneck. This paradigm shift establishes diffusion models not merely as theoretical generators, but as precision tools for targeted medical image analysis and lesion segmentation (Wu et al., 2022).

2.3 Network Architectures

To implement the training objective $L_{simple}(\theta)$ established in Definition 3, the noise predictor $\epsilon_\theta(x_t, t)$ must be parameterised by a neural network capable of processing both the latent state x_t and the discrete temporal index t (Ho et al., 2020). Since the network parameters θ are shared across all discrete timesteps $t \in \{1, \dots, T\}$, a temporal encoding mechanism $\gamma(t)$ is required to break the permutation symmetry of the time index (Ho et al., 2020; Vaswani et al., 2017). Furthermore, the selection of the network topology is dictated by the dimensionality d and the spatial inductive bias of the target data manifold x_0.

Definition 9 (Sinusoidal Time Embedding Vaswani et al. (2017); Ho et al. (2020))**.** For each discrete timestep $t \in \{1, \dots, T\}$, the time embedding is defined by the mapping $\gamma : \{1, \dots, T\} \to \mathbb{R}^{d_{emb}}$, where the components for dimension index $i \in \{0, \dots, \frac{d_{emb}}{2} - 1\}$ are given by:

$$\gamma(t)_{2i} = \sin\left(\frac{t}{\omega^{2i/d_{emb}}}\right), \quad \gamma(t)_{2i+1} = \cos\left(\frac{t}{\omega^{2i/d_{emb}}}\right)$$

where $\omega = 10000$ is a pre-defined base frequency constant.

For low-dimensional synthetic datasets (e.g., $x_t \in \mathbb{R}^d$ where $d = 2$), the manifold lacks complex spatial hierarchies or local pixel-wise correlations. In such cases, a Multi-Layer Perceptron (MLP) is utilised to approximate the denoising mapping. To distinguish this specific implementation from the foundational framework, the MLP is parameterised by a distinct weight set $\theta_M \in \Theta_M$.

Definition 10 (MLP-based Noise Predictor Sohl-Dickstein et al. (2015); Ho et al. (2020))**.** For a state $x_t \in \mathbb{R}^d$ and a time embedding $\gamma(t) \in \mathbb{R}^{d_{emb}}$, the MLP-based predictor $\epsilon_{\theta_M} : \mathbb{R}^d \times \mathbb{R}^{d_{emb}} \to \mathbb{R}^d$ is defined as the composition of L hidden layers and a terminal reconstruction layer. The hidden states $h^{(l)} \in \mathbb{R}^{d_h}$ are computed recursively for $l \in \{1, \dots, L\}$ as:

$$h^{(l)} = \sigma\left(W^{(l)} h^{(l-1)} + b^{(l)} + \Phi^{(l)}(\gamma(t))\right)$$

where $h^{(0)} = x_t$, $W^{(l)} \in \mathbb{R}^{d_h \times d_h}$, and $b^{(l)} \in \mathbb{R}^{d_h}$ are learnable parameters. The final noise prediction is obtained via the linear projection:

$$\epsilon_{\theta_M}(x_t, t) = W_{out} h^{(L)} + b_{out}$$

where $W_{out} \in \mathbb{R}^{d \times d_h}$ and $b_{out} \in \mathbb{R}^d$ denote the output layer weights and biases, respectively.

Conversely, for high-dimensional structured data such as images $x_t \in \mathbb{R}^{C \times H \times W}$, a U-Net architecture is employed to preserve spatial covariance and multi-scale

topological features. To ensure rigorous namespace isolation, this network is parameterised by its own unique weight set $\theta_U \in \Theta_U$. To address the information bottleneck inherent in deep convolutional architectures, this structure utilises skip-connections $\oplus$ to fuse high-frequency spatial details from the encoder directly into the decoder (Ronneberger et al., 2015; Ho et al., 2020).

Definition 11 (U-Net Architecture Ronneberger et al. (2015); Ho et al. (2020))**.** The noise predictor $\epsilon_{\theta_U} : \mathbb{R}^{C\times H\times W} \times \mathbb{R}^{d_{emb}} \to \mathbb{R}^{C\times H\times W}$ is defined as a symmetric encoder-decoder. For a given state x_t and time embedding $\gamma(t) \in \mathbb{R}^{d_{emb}}$, let $h_{enc}^{(l)}$ and $h_{dec}^{(l)}$ denote the hidden spatial features at layer $l \in \{1, \dots, L\}$. The internal transitions are governed by the recursive relations:

$$h_{enc}^{(l)} = \mathcal{F}_{down}^{(l)}\big(h_{enc}^{(l-1)}, \gamma(t)\big)$$

$$h_{dec}^{(l)} = \mathcal{F}_{up}^{(l)}\big(h_{dec}^{(l+1)} \oplus h_{enc}^{(l)}, \gamma(t)\big)$$

where $h_{enc}^{(0)} = x_t$, $\mathcal{F}_{down}^{(l)}$ and $\mathcal{F}_{up}^{(l)}$ are layer-wise convolutional operators, and $\oplus$ signifies channel-wise concatenation. The final noise prediction is defined by the terminal reconstruction mapping:

$$\epsilon_{\theta_U}(x_t, t) = \mathcal{F}_{out}(h_{dec}^{(1)})$$

where $\mathcal{F}_{out}$ is a 1×1 convolutional operator that projects the decoded features back to the target dimensionality $C \times H \times W$.

2.4 Experimental Metrics

To quantitatively evaluate the alignment between the true data distribution $q_0(x_0)$ and the generative distribution $p_\theta(x_0)$, the selection of metrics is strictly dictated by the dimensionality of the state space. For low-dimensional continuous manifolds (e.g., $x_0 \in \mathbb{R}^d$), geometric overlap and statistical distances are evaluated directly in the original data space.

Definition 12 (Empirical 2-way Coverage)**.** Given a reference set of real samples $\mathcal{S}_{ref} = \{x_0^{(1)}, \dots, x_0^{(N)}\} \sim q_0(x_0)$ and a generated set $\mathcal{S}_{gen} = \{y_0^{(1)}, \dots, y_0^{(M)}\} \sim p_\theta(x_0)$ where $x_0^{(i)}, y_0^{(j)} \in \mathbb{R}^d$. Let $d : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}_{\geq 0}$ be a distance metric and $\tau > 0$ be a predefined proximity threshold. The 2-way Coverage is defined by two directional components:

1. **Recall-based Coverage** ($\mathrm{Cov}_{R\to G}$):

$$\mathrm{Cov}_{R\to G} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left(\min_{y_0\in\mathcal{S}_{gen}} d(x_0^{(i)}, y_0) \leq \tau\right)$$

2. **Precision-based Coverage** ($\mathrm{Cov}_{G\to R}$):

$$\mathrm{Cov}_{G\to R} = \frac{1}{M}\sum_{j=1}^{M} \mathbb{I}\left(\min_{x_0\in\mathcal{S}_{ref}} d(y_0^{(j)}, x_0) \leq \tau\right)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function. The evaluation is reported as the tuple $(\text{Cov}_{R\to G}, \text{Cov}_{G\to R})$.

In Definition 12, $\text{Cov}_{R\to G}$ serves as a measure of **recall**. It verifies whether the various data modes present in the real-world distribution are captured by the generative model, thereby constituting a primary basis for diagnosing mode collapse. Conversely, $\text{Cov}_{G\to R}$ assesses **precision** by verifying whether synthetic samples reside within the τ-neighbourhood of the true data manifold, which evaluates sample fidelity. Compared to unidirectional metrics, the 2-way coverage framework provides a symmetric perspective to evaluate the alignment between the supports of the generative distribution $p_\theta(x_0)$ and the true distribution $q_0(x_0)$.

Definition 13 (Maximum Mean Discrepancy Kanamori et al. (2010); Gretton et al. (2012))**.** For the empirical sets $\mathcal{S}_{ref}$ and $\mathcal{S}_{gen}$ defined in Definition 12, given a symmetric positive-definite kernel function $k : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$, the unbiased empirical estimator for the squared MMD is defined as:

$$\widehat{\text{MMD}}^2 = \frac{1}{N(N-1)} \sum_{i\neq j}^{N} k(x_0^{(i)}, x_0^{(j)}) + \frac{1}{M(M-1)} \sum_{i\neq j}^{M} k(y_0^{(i)}, y_0^{(j)}) - \frac{2}{NM} \sum_{i=1}^{N} \sum_{j=1}^{M} k(x_0^{(i)}, y_0^{(j)})$$

where k is typically implemented as the Gaussian RBF kernel $k(u, v) = \exp(-\|u - v\|^2/2\sigma^2)$.

The primary advantage of $\widehat{\text{MMD}}^2$ in Definition 13 is that it obviates the requirement for explicit probability density estimation of $q_0(x_0)$ or $p_\theta(x_0)$ in high-dimensional spaces. By employing the kernel trick, it maps distributions into a Reproducing Kernel Hilbert Space (RKHS), evaluating the functional alignment between $p_\theta(x_0)$ and $q_0(x_0)$ through intra-set cohesion and inter-set repulsion. An $\widehat{\text{MMD}}^2$ value approaching zero indicates both high sample fidelity and comprehensive mode coverage.

Conversely, for high-dimensional structured data such as images $x_0 \in \mathbb{R}^{C\times H\times W}$, direct distance computations in the pixel space fail to meaningfully capture semantic similarities. Therefore, evaluations are conducted in a latent feature space $\mathbb{R}^{d_{feat}}$ extracted via a pre-trained classification network $\phi : \mathbb{R}^{C\times H\times W} \to \mathbb{R}^{d_{feat}}$ (typically the Inception-v3 architecture).

Definition 14 (Fréchet Inception Distance Heusel et al. (2017))**.** Let $\phi(\mathcal{S}_{ref})$ and $\phi(\mathcal{S}_{gen})$ denote the sets of extracted features in $\mathbb{R}^{d_{feat}}$. Let $\mu_r, \mu_g \in \mathbb{R}^{d_{feat}}$ denote their respective empirical mean vectors, and $\Sigma_r, \Sigma_g \in \mathbb{R}^{d_{feat}\times d_{feat}}$ denote their empirical covariance matrices. The FID is defined as the Fréchet distance (Wasserstein-2 distance) between these two multivariate Gaussian approximations:

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\right)$$

where $\text{Tr}(\cdot)$ denotes the trace operator of a matrix.

Definition 15 (Feature-based Precision and Recall Kynkäänniemi et al. (2019))**.** Operating within the feature space $\mathbb{R}^{d_{feat}}$, let $d(\cdot,\cdot)$ be the Euclidean distance. Define $B(\phi(x_0), k)$ as the distance from a feature vector $\phi(x_0)$ to its k-th nearest neighbour within its own respective set. The Feature-based Precision and Recall are defined as:

$$\text{Precision} = \frac{1}{M}\sum_{j=1}^{M} \mathbb{I}\left(\exists x_0 \in \mathcal{S}_{ref} : d(\phi(y_0^{(j)}), \phi(x_0)) \leq B(\phi(x_0), k)\right)$$

$$\text{Recall} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left(\exists y_0 \in \mathcal{S}_{gen} : d(\phi(x_0^{(i)}), \phi(y_0)) \leq B(\phi(y_0), k)\right)$$

The metrics established in Definition 14 and Definition 15 effectively extend the evaluation principles of low-dimensional manifolds into high-dimensional semantic spaces. The FID provides a macroscopic measure of distribution alignment by calculating the Wasserstein-2 distance between the multivariate Gaussian approximations $\mathcal{N}(\mu_r, \Sigma_r)$ and $\mathcal{N}(\mu_g, \Sigma_g)$ of the extracted features $\phi(x_0) \in \mathbb{R}^{d_{feat}}$; a lower FID indicates a closer overall agreement between the generative distribution $p_\theta(x_0)$ and the true data distribution $q_0(x_0)$.

However, because the FID conflates sample fidelity and diversity into a single scalar, Feature-based Precision and Recall are utilised to explicitly disentangle these properties. Analogous to the low-dimensional 2-way Coverage framework ($\text{Cov}_{R\to G}, \text{Cov}_{G\to R}$), Feature Precision quantifies the realism of the generated samples $y_0 \sim p_\theta(x_0)$ within the latent space, whilst Feature Recall evaluates the proportion of the true data manifold $x_0 \sim q_0(x_0)$ that is accurately captured. Together, they provide a rigorous and symmetric diagnostic tool for identifying mode collapse in complex image synthesis.

3 Methodology

3.1 Network Architecture Design

The multi-layer perceptron (MLP) was used for the lower-dimensional datasets like the 2D manifolds (see Section 4.1.1). The MLP had 4 hidden layers with 64 neurons each. The timestep was stacked with each variable. This setup had 17,000 total parameters.

U-Net was used for higher-dimensional data with a spatial component, we used a U-Net with 2 down-sampling channels, 2 up-sampling channels, and a bottleneck. Each channel consisted of 2 convolutional layers and a max-pooling layer for halving the height and width. The feature maps used were $\{64, 128, 256, 128, 64\}$, where the bottleneck component used 256 feature maps. The total number of parameters was 4 million. The time was embedded using a linear layer to project the time from a scalar to a vector of size $\{64, 128, 256\}$,

so the time embedding vector could be broadcast and added to all the feature maps at each channel.

3.2 Noise Schedule Implementation

The variance scheduler determines how the variance of the injected noise changes across the timesteps. The linear schedule results in a very steep decrease in image quality around $0.25T$, whereas the cosine schedule offers a gradual noising of the clean image, leading to the diffusion model not biasing any specific timestep. However, the linear schedule is sometimes preferred due to its simplicity of implementation.

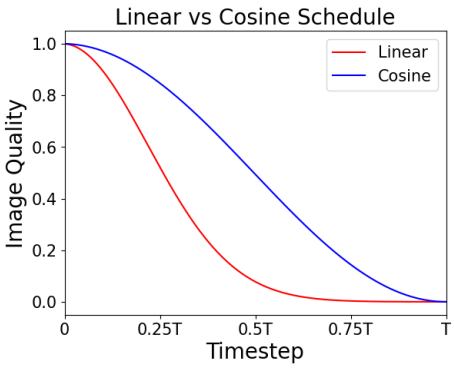


Figure 1: Image quality from clean (1.0) to Gaussian noise (0.0)

The linear schedule (Ho et al., 2020) we used was linearly increased from 0.0001 to 0.02 to fit the total diffusion steps $T = 500$.

$$\beta_t = 0.0001 + \frac{t-1}{T-1}0.02$$

The cosine schedule (Nichol and Dhariwal, 2021) directly parametrises the $\bar{\alpha}'_t$, using a small offset of 0.008 so the variance $\beta'_t = 1 - \frac{\bar{\alpha}'_t}{\bar{\alpha}'_{t-1}}$ is not too small near $t = 0$.

$$\bar{\alpha}'_t = \cos^2\left(\frac{\frac{t}{T} + 0.008}{1 + 0.008} \cdot \frac{\pi}{2}\right)$$

3.3 Data Preparation Preprocessing

3.3.1 Normalisation: zero-mean scaling

Scaling the data is an important part of preprocessing for performance and future-proofing. Most image datasets represent images as integers between 0 and 255, but others could use a different scale; therefore, training all the models with a consistent scale will be good for future-proofing. All the data we used

was normalised by subtracting the dataset mean and dividing by the dataset standard deviation. Normalising the dataset such that the tensors have a mean of 0 can reduce the effects of high divergence at the beginning of training for very large models.

3.4 Experimental Environment and Configuration

3.4.1 Training Optimiser

The optimiser is used to update the model parameters using the calculated gradients from back-propagation. The simplest optimiser is stochastic gradient descent (SGD), which just updates the parameters based on the batch of samples, $\theta = \theta - \text{LR} \cdot \nabla\theta$. However, for the experiments, we used the Adam optimiser (Kingma and Ba, 2014), which can decrease the training time. Adam dynamically changes the step-size to account for momentum and the gradient computed from the loss function being too small or too large.

3.4.2 Training Parameters

For the toy datasets, each sample had only 2 dimensions, so a batch size of 2048 was used; training for 100 epochs was sufficient to generate the samples of the data distribution. For the higher-dimensional datasets, we used a batch size of 64 due to the high amount of GPU memory being used. The model was also trained for 100 epochs and took around 20 minutes to start creating good-quality images.

4 Experiments

In our experiments, we aim to validate the generation performance of a standard Denoising Diffusion Probabilistic Model (DDPM). Specifically, our main focus is to analyse the model's capacity to accurately learn and reconstruct complex data distributions, ranging from low-dimensional continuous manifolds to higher-dimensional image spaces.

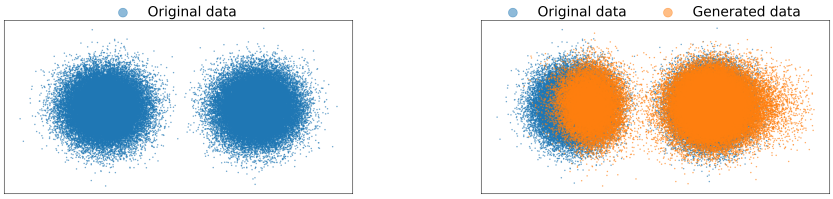
Datasets and Metrics. For this evaluation, we employ two datasets: a 2D Gaussian mixture toy dataset (sampled from a normal distribution with standard deviation $\sigma = 0.5$) and the MNIST dataset (28×28, standard 60,000 training split). To assess generation quality on the 2D dataset, we report $\text{Cov}_{G \to R}$, $\text{Cov}_{R \to G}$, and MMD^2. For the MNIST dataset, we report the Fréchet Inception Distance (FID), alongside Precision and Recall.

The remainder of this section is organized as follows. Section 4.1 presents the main generation results on both datasets. Section 4.2 then details ablation studies conducted on the MNIST dataset, specifically exploring the performance trade-offs associated with the total diffusion timesteps (T) and investigating the underlying dynamics of different noise variance schedules (Linear versus Cosine).

4.1 Main Results

4.1.1 Toy Dataset

We first evaluate the DDPM on the 2D Gaussian dataset. Visually, the model successfully captures the original bi-modal structure (Figure 2), with the denoising trajectory systematically steering noise into target clusters (Figure 3). Quantitatively, Table 1 confirms this capability, reporting balanced coverage metrics ($\text{Cov}_{R\to G} = 0.713$, $\text{Cov}_{G\to R} = 0.779$) and a low distributional distance ($\text{MMD}^2 = 0.054$). Together, these results explicitly validate the capacity of the DDPM to learn and reconstruct data distributions on the 2D toy dataset.



(a) Ground Truth (b) Generated

Figure 2: Qualitative comparison of real and generated distributions on the 2D Gaussian dataset. Blue dots represent the ground truth, while orange dots denote DDPM-generated samples. The generated samples accurately capture the original bi-modal structure, demonstrating effective distribution matching.

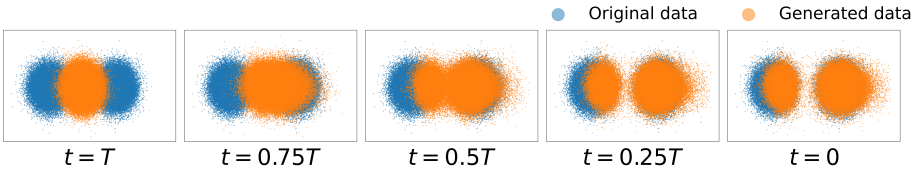


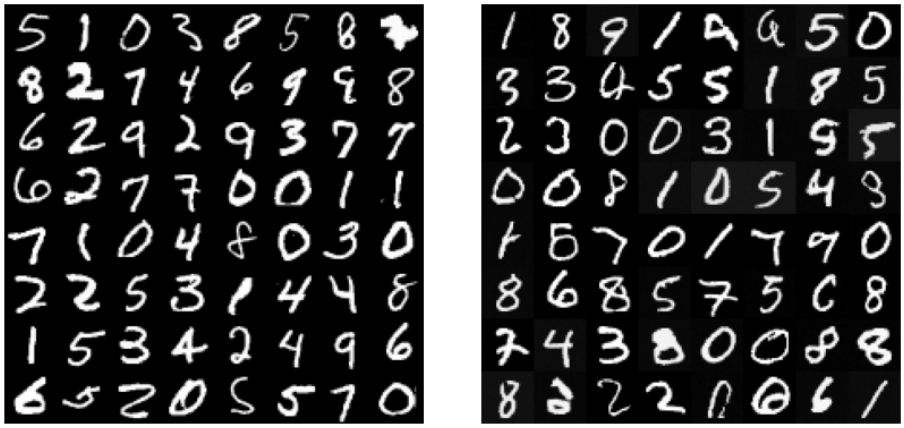
Figure 3: The reverse diffusion (denoising) trajectory on the 2D Gaussian dataset from $t = T$ to $t = 0$. Snapshots at intermediate timesteps track the spatial evolution of the generated data relative to the original distribution. As the process progresses, the DDPM effectively steers the initially scattered noise to reconstruct the target bi-modal structure.

Table 1: Quantitative evaluation of generation quality on the 2D Gaussian mixture dataset. Metrics are reported as Mean (± Std) computed over 10 independent sampling runs ($N = 10000$ per run). The results establish the baseline performance of the DDPM, demonstrating its capacity to balance recall ($\text{Cov}_{R\to G}$) and precision ($\text{Cov}_{G\to R}$) while maintaining a low distributional distance (MMD^2).

Dataset	**Cov**$_{R\to G}$ ↑	**Cov**$_{G\to R}$ ↑	**MMD**^{2} ↓
2D Gaussian Mixture	0.713 (0.003)	0.779 (0.004)	0.054 (0.001)

4.1.2 Image Dataset

We next evaluate the DDPM on the MNIST dataset to verify its generation capabilities in a higher-dimensional image space. Visually, the model successfully replicates the distinct character strokes and structural diversity of the original handwritten digits (Figure 4), with the denoising trajectory illustrating the gradual emergence of clear structures from pure noise (Figure 5). Quantitatively, as reported in Table 2, the model achieves a remarkably low FID of 1.72. Furthermore, the high Recall score (0.901) combined with a solid Precision score (0.827) demonstrates that the model effectively captures the full diversity of the data distribution without suffering from mode collapse. These results confirm the robustness of the DDPM for high-fidelity image generation.



(a) Ground Truth (b) Generated

Figure 4: Qualitative comparison of real and generated samples on the MNIST dataset. The left panel displays the ground truth handwritten digits, while the right panel shows the digits generated by the DDPM. Visually, the model successfully replicates the structural diversity and distinct character strokes of the original data.

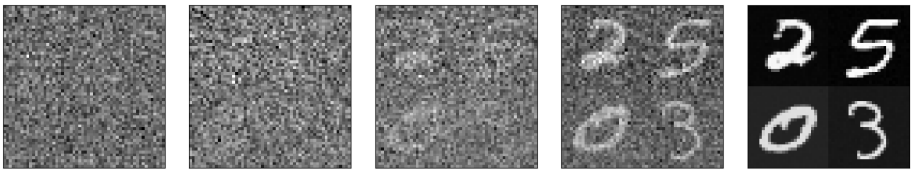


Figure 5: Visualization of the reverse diffusion process on MNIST under DDPM. This sequence shows the progression from pure noise (x_T) to the final generated digits (x_0). As the denoising steps proceed, clear and recognizable digit structures gradually emerge, demonstrating the model's ability to effectively reconstruct complex high-dimensional image data.

Table 2: Quantitative evaluation of image generation performance on the MNIST dataset. Metrics are presented as Mean (± Std) across 10 independent runs ($N = 5000$). The remarkably low FID and high Recall score demonstrate the model's ability to generate high-fidelity samples while capturing full data diversity without mode collapse.

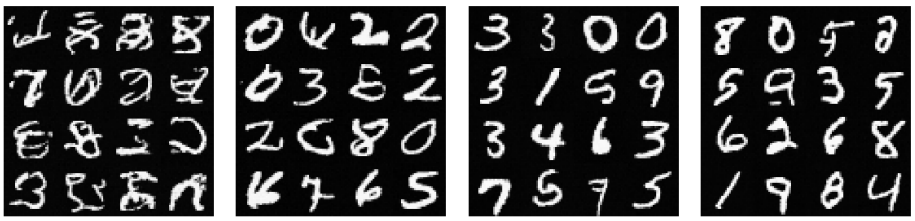
Dataset	**Precision ↑**	**Recall ↑**	**FID ↓**
MNIST numbers	0.694 (0.01)	0.674 (0.01)	6.13 (0.07)

4.2 Ablation studies on MNIST

In this section, we conduct ablation studies on the MNIST dataset to isolate and evaluate the impact of critical hyperparameters on the generative performance of the DDPM framework. Specifically, we investigate two core components: the total number of diffusion timesteps (T) and the choice of the forward noise schedule. By analysing both quantitative metrics and qualitative visual generations, we aim to identify the optimal configuration that balances computational efficiency with generation fidelity, while further exploring the structural characteristics of the baseline model.

4.2.1 Impact of diffusion timesteps T

This section evaluates the influence of total diffusion timesteps (T). Visually, Figure 6 shows that structural clarity improves significantly as T increases from 100 to 500. Quantitatively, Table 3 confirms this trend, with FID dropping from 20.1 to a minimal 6.13 at $T = 500$, before plateauing at $T = 1000$. Theoretically, according to Nichol and Dhariwal (2021), higher timesteps reduce discretization errors, allowing the reverse process to better approximate the continuous diffusion trajectory. However, beyond $T = 500$, the accumulation of sampling noise offsets further structural gains. Thus, $T = 500$ offers the optimal balance between generative fidelity and computational efficiency.



(a) T=100 (b) T=250 (c) T=500 (d) T=1000

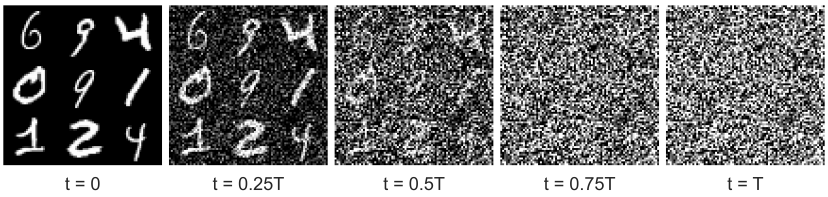
Figure 6: Qualitative comparison of generated MNIST digits under varying total diffusion timesteps (T). The four subfigures illustrate the evolution of sampled digit structures as the number of reverse discretisation steps increases from $T = 100$ to $T = 1000$. Visual fidelity and structural clarity improve significantly up to $T = 500$, after which the generation quality plateaus, demonstrating that $T = 500$ provides the optimal balance between reducing discretization errors and maintaining sampling efficiency.

Table 3: Quantitative impact of total diffusion timesteps (T) on MNIST image generation performance. Metrics are presented as Mean ($\pm$ Std) across 10 independent runs ($N = 5000$), with the best practical results (excluding the Real Data Baseline) highlighted in bold. The data demonstrates that increasing timesteps significantly improves generation quality and diversity up to $T = 500$, after which the performance plateaus, indicating diminishing returns.

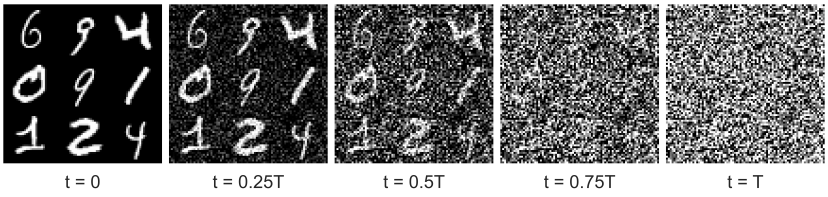
Timesteps T	**Precision ↑**	**Recall ↑**	**FID ↓**
100	0.701 (0.02)	0.150 (0.01)	20.1 (0.21)
250	**0.710** (0.01)	0.589 (0.01)	11.8 (0.01)
500	0.694 (0.01)	**0.674** (0.01)	**6.13** (0.07)
1000	0.682 (0.01)	0.668 (0.02)	6.45 (0.08)

4.2.2 Impact of noise schedule methods

This section investigates the impact of the forward variance schedule. Visually, Figure 7 shows that the linear schedule rapidly corrupts MNIST digits, whereas the cosine schedule introduces noise more gradually. Despite this, quantitative gains are marginal (Table 4): FID improves slightly to 6.133, while Precision and Recall remain unchanged. Theoretically, this aligns with Nichol and Dhariwal (2021) that DDPM's fixed variance parameterisation restricts it from fully exploiting the smoothed noise trajectory, leaving generative improvements structurally constrained.



(a) Linear schedule



(b) Cosine schedule

Figure 7: Visual progression of MNIST digit samples along the forward diffusion process, comparing linear and cosine noise schedules. Subfigures display images from $t = 0$ to $t = T$ at distinct time points, illustrating noise accumulation rates. Compared to the linear schedule, the cosine schedule introduces noise more gradually.

Table 4: Quantitative comparison of DDPM performance trained with linear and cosine variance schedules. Metrics are presented as Mean ($\pm$ Std) across 10 independent runs ($N = 5000$), with the best practical results (excluding the Real Data Baseline) highlighted in bold. While the cosine schedule yields a 10% improvement in FID (6.133 vs. 6.721), the overall metrics remain comparable, indicating that adjusting the schedule alone provides limited quantitative gains on simple datasets.

Variance Schedule	Precision ↑	Recall ↑	FID ↓
Linear	0.691 (0.006)	**0.678** (0.007)	6.721 (0.131)
Cosine	**0.694** (0.012)	0.674 (0.011)	**6.133** (0.071)

5 Discussion and Conclusion

In this report, our experimental results demonstrate the effectiveness of the standard DDPM in generation tasks. Theoretically, its generative capability relies heavily on the assumption of single-step Gaussian transitions within a continuous Markov chain (Ho et al., 2020). Our ablation studies in Section 4.2.1 validated this premise: given a fixed total noise scale, a model trained with $T = 100$ performed worse than one with $T = 500$. This is because to ensure the reverse process remains approximately Gaussian, DDPM must employ extremely small noise increments. Consequently, when processing complex natural

images such as CIFAR-10, this mathematical condition requires introducing additional generation steps—typically up to 1000 discrete timesteps—leading to a substantial increase in computational overhead (Ho et al., 2020).

Accelerating the Sampling Process. To address this computational bottleneck, variants such as Improved DDPM and DDIM both focus on refining the Gaussian assumption. Specifically, when a reduction in sampling steps increases the single-step variance and compromises the Gaussian assumption, Improved DDPM directly parameterises the reverse variance and introduces a cosine schedule to mitigate rapid information loss at reduced steps (Nichol and Dhariwal, 2021). However, our Section 4.2.2 results revealed that applying the cosine schedule in isolation yielded only marginal gains. This finding corroborates Improved DDPM's core premise: the cosine schedule must be coupled with variance parameterisation to fully realise its generative benefits. Alternatively, DDIM maintains the same marginal Gaussian distributions while constructing a non-Markovian forward process (Song et al., 2021). This enables sampling acceleration simply by modifying the sampling strategy without retraining the base model. Together, these two variants provide a theoretical foundation for applying the DDPM framework to more complex datasets.

Extending to Discrete Modalities. In addition to computational bottlenecks, the standard DDPM's reliance on the Gaussian assumption also restricts the breadth of applicable data modalities. For instance, in the MNIST image generation task, the model must normalise pixel values into a continuous range during data preprocessing (See Section 3.3.1) to satisfy the mathematical prerequisite of additive continuous Gaussian noise. Conversely, discrete data, such as text or categorical medical labels, cannot be mapped to a continuous state space through simple normalisation techniques. To overcome this limitation, D3PM reformulates the forward process using categorical transition matrices and conceptualises the traditional addition of continuous Gaussian noise as probabilistic state transitions between discrete categories (Austin et al., 2021). By doing so, D3PM establishes the theoretical foundation for extending diffusion models to non-continuous data distributions.

Transitioning to Controllable Generation. Beyond the constraints regarding sampling efficiency and discrete data modalities, another critical limitation of the standard DDPM lies in its uncontrollable generation process. Fundamentally, the standard DDPM operates as an unconditional generative model optimised to approximate the marginal data distribution, $q_0(x_0)$. In practice—particularly within clinical medical imaging—there is a strict demand for controllable synthesis driven by specific conditions, requiring the modelling of the conditional distribution, $q_0(x_0 \mid y)$. To achieve this transition, conditional information must be explicitly injected into either the sampling or the training phase. During the sampling phase, Classifier Guidance freezes the base DDPM and steers the denoising trajectory using gradients from an auxiliary classifier (Dhariwal and Nichol, 2021). Conversely, Classifier-Free Guidance injects

the specific labels directly into the pre-training phase, enabling the diffusion model to jointly learn the conditional distribution $q_t(x_t \mid y)$ and the unconditional distribution $q_t(x_t)$ within a unified model (Ho and Salimans, 2021). These mechanisms establish the theoretical foundation for downstream conditional applications. Following this training-based paradigm, frameworks like MedSegDiff incorporate medical scans as explicit conditions during the diffusion process, guiding precise medical image segmentation without relying on external classifiers (Wu et al., 2022).

Future Work. Given the practical importance of conditional guidance and inspired by how frameworks like MedSegDiff integrate guidance mechanisms with foundational diffusion models, we pose the following question: Can we incorporate guidance strategies into the baseline DDPM established in this report to achieve conditional generation? In our subsequent work, we will further explore the underlying principles of conditional diffusion models, aiming ultimately to realise highly precise and controllable image synthesis in complex scenarios.

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A Detailed Mathematical Proofs

A.1 Proof of Proposition 1

Proof of Proposition 1. According to the recursive relationship in Definition 1, we have $x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{1-\alpha_t}\epsilon_t$. Expanding this backward by one step and substituting $x_{t-1} = \sqrt{\alpha_{t-1}}x_{t-2} + \sqrt{1-\alpha_{t-1}}\epsilon_{t-1}$ yields:

$$x_t = \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{\alpha_t(1-\alpha_{t-1})}\epsilon_{t-1} + \sqrt{1-\alpha_t}\epsilon_t$$

Since ϵ_{t-1} and ϵ_t are independently sampled standard Gaussian noises, by the additive property of independent normal distributions, their linear combination is equivalent to a new Gaussian noise with a variance equal to the sum of their respective variances:

$$\alpha_t(1-\alpha_{t-1}) + (1-\alpha_t) = 1 - \alpha_t\alpha_{t-1}$$

Thus, we can rewrite the equation as

$$x_t = \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\bar{\epsilon}_{t-2}$$

where $\bar{\epsilon}_{t-2} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. By continuously expanding this process to the initial state x_0 via mathematical induction, the product of the mean coefficients converges to $\sqrt{\bar{\alpha}_t}$, and the accumulated variance converges to $1-\bar{\alpha}_t$, which completes the proof. □

A.2 Proof of Proposition 2

Proof of Proposition 2. According to Bayes' theorem, the conditional posterior distribution $q_t(x_{t-1} \mid x_t, x_0)$ can be expressed as:

$$q_t(x_{t-1} \mid x_t, x_0) = q_t(x_t \mid x_{t-1}, x_0)\frac{q_t(x_{t-1} \mid x_0)}{q_t(x_t \mid x_0)}$$

Since the forward process is a Markov chain, $q_t(x_t \mid x_{t-1}, x_0)$ reduces to $q_t(x_t \mid x_{t-1})$ (Ho et al., 2020). Based on Definition 1 and Proposition 1, we substitute the following Gaussian densities:

- $q_t(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, \beta_t\mathbf{I})$
- $q_t(x_{t-1} \mid x_0) = \mathcal{N}(x_{t-1}; \sqrt{\bar{\alpha}_{t-1}}x_0, (1-\bar{\alpha}_{t-1})\mathbf{I})$
- $q_t(x_t \mid x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)\mathbf{I})$

Ignoring terms that do not involve x_{t-1}, the probability density is proportional to:

$$\begin{aligned} q_t(x_{t-1} \mid x_t, x_0) &\propto \exp\left(-\frac{1}{2}\left(\frac{(x_t - \sqrt{\alpha_t}x_{t-1})^2}{\beta_t} + \frac{(x_{t-1} - \sqrt{\bar{\alpha}_{t-1}}x_0)^2}{1-\bar{\alpha}_{t-1}}\right)\right) \\ &= \exp\left(-\frac{1}{2}\left(\frac{x_t^2 - 2\sqrt{\alpha_t}x_t x_{t-1} + \alpha_t x_{t-1}^2}{\beta_t} + \frac{x_{t-1}^2 - 2\sqrt{\bar{\alpha}_{t-1}}x_0 x_{t-1} + \bar{\alpha}_{t-1}x_0^2}{1-\bar{\alpha}_{t-1}}\right)\right) \end{aligned}$$

Combining the terms related to x_{t-1}^2 and x_{t-1}, we identify the coefficients for a standard Gaussian form $\exp(-\frac{1}{2}(\frac{1}{\tilde{\beta}_t}x_{t-1}^2 - \frac{2\tilde{\mu}_t}{\tilde{\beta}_t}x_{t-1} + C))$. The inverse variance $\frac{1}{\tilde{\beta}_t}$ is:

$$\frac{1}{\tilde{\beta}_t} = \frac{\alpha_t}{\beta_t} + \frac{1}{1-\bar{\alpha}_{t-1}} = \frac{\alpha_t(1-\bar{\alpha}_{t-1}) + \beta_t}{\beta_t(1-\bar{\alpha}_{t-1})}$$

Utilising the identity $\alpha_t = 1 - \beta_t$, the numerator simplifies to:

$$(1-\beta_t)(1-\bar{\alpha}_{t-1}) + \beta_t = 1 - \bar{\alpha}_{t-1} - \beta_t + \bar{\alpha}_t + \beta_t = 1 - \bar{\alpha}_t$$

Thus, the posterior variance is $\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$. Next, for the posterior mean $\tilde{\mu}_t$:

$$\begin{aligned}
\frac{\tilde{\mu}_t}{\tilde{\beta}_t} &= \frac{\sqrt{\alpha_t}}{\beta_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}x_0 \\
\tilde{\mu}_t &= \tilde{\beta}_t\left(\frac{\sqrt{\alpha_t}}{\beta_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}x_0\right) \\
&= \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t\left(\frac{\sqrt{\alpha_t}}{\beta_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}x_0\right)
\end{aligned}$$

Distributing the terms yields the final expression:

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}x_0$$

This completes the proof. □

A.3 Proof of Proposition 3

Proof of Proposition 3. From Proposition 2, the true conditional posterior mean is analytically defined as:

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}x_0$$

Recall from the forward marginal distribution (Proposition 1) that the state x_t can be directly sampled from x_0 using the reparameterisation trick:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$$

where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Rearranging this equation to solve for the initial data state x_0 yields:

$$x_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(x_t - \sqrt{1-\bar{\alpha}_t}\epsilon\right)$$

Substituting this expression for x_0 into the analytical form of $\tilde{\mu}_t$ above, we obtain:

$$\tilde{\mu}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\left(\frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \sqrt{1-\bar{\alpha}_t}\epsilon)\right) + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t$$

Recognising that $\bar{\alpha}_t = \alpha_t \bar{\alpha}_{t-1}$, we can simplify the fraction preceding the parenthesis:

$$\frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{(1-\bar{\alpha}_t)\sqrt{\alpha_t \bar{\alpha}_{t-1}}} = \frac{\beta_t}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}}$$

Expanding the terms, the expression becomes:

$$\tilde{\mu}_t = \frac{\beta_t}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}} x_t - \frac{\beta_t \sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}} \epsilon + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} x_t$$

Next, we factor out x_t and combine its coefficients:

$$\begin{aligned} \text{Coefficient of } x_t &= \frac{\beta_t + \alpha_t(1-\bar{\alpha}_{t-1})}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}} \\ &= \frac{\beta_t + \alpha_t - \alpha_t \bar{\alpha}_{t-1}}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}} \end{aligned}$$

Since $\beta_t + \alpha_t = 1$ by definition, the numerator simplifies to $1 - \alpha_t \bar{\alpha}_{t-1} = 1 - \bar{\alpha}_t$. Thus, the coefficient of x_t strictly reduces to:

$$\frac{1-\bar{\alpha}_t}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}} = \frac{1}{\sqrt{\alpha_t}}$$

Similarly, simplifying the coefficient of the ϵ term yields:

$$-\frac{\beta_t \sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}} = -\frac{\beta_t}{\sqrt{\alpha_t}\sqrt{1-\bar{\alpha}_t}}$$

Combining both simplified terms, we arrive at the final analytical form:

$$\tilde{\mu}_t(x_t, \epsilon) = \frac{1}{\sqrt{\alpha_t}} x_t - \frac{\beta_t}{\sqrt{\alpha_t}\sqrt{1-\bar{\alpha}_t}} \epsilon = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon \right)$$

This completes the proof. □

A.4 Proof of Proposition 4

Proof of Proposition 4. Recall the intermediate transition loss L_{t-1} formulated in Theorem 1, which aligns the true posterior mean and the parameterised reverse mean:

$$L_{t-1} = \mathbb{E}_{q_t} \left[\frac{1}{2\sigma_t^2} \|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\|^2 \right] + C$$

From Proposition 3, we substitute the reparameterised expressions for both the true conditional posterior mean $\tilde{\mu}_t(x_t, \epsilon)$ and the model's predicted mean $\mu_\theta(x_t, t)$:

$$\tilde{\mu}_t(x_t, \epsilon) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon \right)$$

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right)$$

Calculating the difference between these two mean vectors inside the L_2 norm yields:

$$\begin{aligned}
\tilde{\mu}_t(x_t, \epsilon) - \mu_\theta(x_t, t) &= \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon \right) - \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) \\
&= \frac{1}{\sqrt{\alpha_t}} \left(\frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon \right) \\
&= \frac{\beta_t}{\sqrt{\alpha_t}\sqrt{1-\bar{\alpha}_t}} \left(\epsilon_\theta(x_t, t) - \epsilon \right)
\end{aligned}$$

Since the L_2 norm is symmetric with respect to its arguments (i.e., $\|a-b\|^2 = \|b-a\|^2$), we can factor out the scalar coefficient and express the squared norm as:

$$\|\tilde{\mu}_t(x_t, \epsilon) - \mu_\theta(x_t, t)\|^2 = \left(\frac{\beta_t}{\sqrt{\alpha_t}\sqrt{1-\bar{\alpha}_t}} \right)^2 \|\epsilon - \epsilon_\theta(x_t, t)\|^2$$

Squaring the scalar coefficient simplifies to:

$$\left(\frac{\beta_t}{\sqrt{\alpha_t}\sqrt{1-\bar{\alpha}_t}} \right)^2 = \frac{\beta_t^2}{\alpha_t(1-\bar{\alpha}_t)}$$

Substituting this squared difference back into the original expectation for L_{t-1}, and replacing the expectation over q_t with the explicit expectation over the initial data x_0 and the standard Gaussian noise ϵ, we obtain the final equivalent objective:

$$L_{t-1} = \mathbb{E}_{x_0, \epsilon} \left[\frac{\beta_t^2}{2\sigma_t^2 \alpha_t (1-\bar{\alpha}_t)} \|\epsilon - \epsilon_\theta(x_t, t)\|^2 \right] + C$$

This completes the proof. □